RWANDA POLYTECHNIC

KIGALI COLLEGE

Department of Information and Communication Technology

PROJECT-BASED PRACTICAL TASK

Two Rwandan Scenarios: Build Machine-Learning-Ready Datasets

Item	Details
Competence	10. Apply Machine Learning
Element of competence	1. Apply dataset preparation mechanisms
Performance criteria covered	1.1 Environment configured | 1.2 Data acquired | 1.3 Data pre-processed | 1.4 Features engineered
Mode of work	Groups of two students (pair work). All groups follow the same procedure.
Task format	Question 1 (40 marks): a given scenario from Rwanda with a dataset supplied with this task. Question 2 (60 marks): your own Rwandan scenario, with data you find or collect.
Tools	Anaconda (Python) and Jupyter Notebook
Learning approach	Project-based learning: you do the work hands-on in pairs, take the decisions and justify them
Final product	For each question: a documented, reproducible, model-ready dataset with a pre-processing pipeline. Plus one Git repository, a memo to the manager (Q1), and a documentation report with screenshots of every stage
Presentation	Thursday 12 November 2026, in the morning: 10 minutes per group + 5 minutes of questions
Academic year	2026/2027

1. Purpose of the Task

Machine learning models are only as good as the data used to train them. In this practical task each group of two students first solves a scenario set in Rwanda that is given in this task (Question 1), and then creates and solves a scenario of its own (Question 2). In both questions the group builds the working environment, obtains suitable data, cleans it, engineers features and finishes with a dataset that is ready to be given to a learning algorithm (Element 2).

By the end of the task each student must be able to:

- Configure a reproducible machine learning environment (Anaconda, Jupyter Notebook, libraries, Git).
- Define a problem and derive data requirements from it, then acquire data from legitimate sources.

- Explore, assess and pre-process data: missing values, duplicates, outliers, inconsistent formats, encoding, scaling and splitting.
- Create, transform and select features based on data analysis findings.
- Document every decision so that another person can repeat the work and get the same result.

2. How You Will Work, Tools and Documentation

This is a hands-on task. You learn by doing: each stage starts with a short introduction to the ideas and a worked example on a sample dataset, and then the practical work is yours. You plan, write the code, take the decisions and explain them.

When	What happens
Start of a stage	Short introduction of the concepts and a worked example on a sample dataset
During a stage	You work in your pair on your own data, take decisions, search the documentation and debug your errors
At each checkpoint	You present your evidence and explain your choices; both members must be able to answer
At the end	You submit the dataset, repository and documentation with screenshots, and give your presentation

Pair work rule. Both students do all the tasks. They work as a pair-programming team: one is the Driver (types the code), the other the Navigator (reads, checks, searches). They swap roles at the start of every stage. Each student keeps their own copy of the notebook and must be able to explain any part of it during the checkpoint questions.

Tools you must use

Tool	Use in this task
Anaconda (Navigator and Anaconda Prompt)	Install Python, create and manage the project environment, install libraries
Jupyter Notebook	Write and run all code, and explain your decisions in Markdown cells. Every notebook starts with a Markdown cell showing the group number, the names of both members, the question (Q1 or Q2) and the stage
Python libraries	pandas, numpy, matplotlib, seaborn and scikit-learn (others only if the project needs them)
Git / GitHub or GitLab	Keep the project folder and show the contribution of both members

Documentation guidelines: take screenshots

Your documentation is your proof of work. At every stage you must take screenshots of your own work and place them in the documentation report that you submit. A stage without screenshots is considered not done.

- Use the Windows Snipping Tool (Win + Shift + S) or the screenshot tool of your computer. Do not take photos of the screen with a phone.
- Show the code and its output together in the same screenshot. Zoom in the notebook if needed so that the text is easy to read. Crop out unrelated windows, passwords and personal information.
- Make your identity visible: the notebook name and the first Markdown cell with the group number and member names should appear in the screenshots.
- Put each screenshot in the documentation under the heading of its stage, with a figure number and a caption (for example: Figure 4: Missing values per column in the maize dataset).
- Under each screenshot write one to three sentences: what it shows, what you found, and what you decided.
- Never edit or fake a screenshot. You may be asked to run the same cell again during your presentation.
- Submit one Word or PDF file per group, named GroupXX_ML_Element1_Documentation. Organise it by question and stage: cover page, Question 1 (Stages 1 to 6), Question 2 (Stages 0 to 6), then the appendices (data dictionary, logs, feature table, contribution log).

Minimum screenshots for each stage (repeat the stages 2 to 6 for Question 2):

Stage	Screenshots required
1	Anaconda Navigator with Jupyter Notebook; Anaconda Prompt showing the active environment and the list of environments; notebook output with the version of every library
2	Data requirements in a Markdown cell; the first rows, shape and info of the loaded data; the completed data dictionary
3	Missing values per column; duplicate count; summary statistics; at least five charts, each with your interpretation
4	Every cleaning step with a before and after view (shape, missing counts, unique values); encoding and scaling output; shapes of the train, validation and test sets
5	The new features (first rows showing the new columns); correlation or feature importance output; the final list of selected features
6	Restart and run all (Kernel menu) finishing without errors; the saved final files in the project folder; the final dataset info; the repository page

Presentation

When: Thursday 12 November 2026, in the morning. All work and the documentation must be complete before your presentation.

- Each group has 10 minutes to present both questions, followed by 5 minutes of questions. Both members must speak and answer.

- Present from your Jupyter notebooks. Keep your screenshots as a backup in case the computer fails.
- Cover: the Question 1 scenario and your memo findings; your Question 2 scenario; the main problems you found; your cleaning and feature decisions and why; and the final dataset you deliver.
- Bring a laptop with the notebooks ready to run, and keep a copy of the repository online or on a flash disk.

3. The Two Questions

The practical task has two questions. Both follow the same stages (Section 5). Question 1 trains you on a scenario and a dataset supplied with this task. In Question 2 you transfer what you learned to a scenario that you create yourselves.

	Question 1 (40 marks)	Question 2 (60 marks)
Scenario	Given: maize yield forecasting for farmer cooperatives in Rwanda	Created by your group, set in Rwanda, approved before you start
Data	File supplied with this task (messy, synthetic data)	Found or collected by your group from legitimate sources
Guidance	Guided: the problem and the deliverables are defined for you	Independent: you define the problem, requirements and decisions
Sessions	1 to 3	3 to 6

Question 1: Predicting Maize Yield for Farmer Cooperatives in Rwanda

Case study: *read the whole case carefully before you start. It tells you who you work for, what they need, what the data looks like, and what "finished" means. The organisation and the people are fictional.*

1. Background

Maize is one of the main food crops grown by smallholder farmers in Rwanda. Many farmers grow it in cooperatives so that they can share inputs, storage and transport, and negotiate better prices with buyers. A cooperative that knows early how much its members will harvest can book transport, prepare storage, agree sales with buyers and plan how much fertiliser and improved seed to request for the next season. A cooperative that guesses wrong can lose part of the harvest to poor storage, or sell too cheaply.

The Maize Cooperative Support Unit (MCSU) is a small advisory team that supports 20 maize cooperatives in 14 districts across the Eastern, Southern, Northern and Western Provinces. Rwanda has three agricultural seasons: Season A (about September to January), Season B (about February to June) and Season C (about June to September, mostly marshland and irrigated plots). The MCSU wants to give every cooperative a reliable yield forecast before each harvest.

2. Your role

You have just been hired as a junior data analyst in the MCSU planning and data team, and you work in a pair with another analyst. Your manager, Mr. Habimana, Head of Planning and Data, gives you your first task. The data science team will build the prediction model next week. Your job comes first: make the data ready, and prove that it is ready.

3. How the data was collected

Over three years (2023 to 2025) field officers visited farms in all four provinces, in all three seasons. For each farm they filled in a paper form: location, planting date, farm size, seed variety, fertiliser used, soil tests, pest level, whether the plot was irrigated, the age of the farmer, and, after harvest, the harvest weight and the yield. Rainfall and temperature for the season were added later from the nearest weather station.

The paper forms were typed into spreadsheets by different officers in different district offices, each with their own habits. The spreadsheets were then combined into one file in a hurry. Some forms were incomplete, and nobody has checked the combined file.

From: Mr. Habimana, Head of Planning and Data, MCSU

To: Junior Data Analysts

Subject: Maize yield data: please prepare it before the modelling team starts

Dear team,

Our cooperatives keep asking for a forecast of maize yield before harvest. Next week the data science team starts to build a prediction model, but I do not trust the file we received from the districts. It was assembled quickly and nobody has checked it.

Please do not build the model. Instead I need you to:

- set up a proper working environment, so that anybody in my team can repeat your work;
- check that the file contains what we need to answer the five questions in the attached list;
- find every problem in the data and tell me how serious each one is;
- clean the data and prepare it so that the modelling team can use it directly, without information from the future leaking into the model;
- create useful new information (features) from what we already have;
- send me a one-page memo in simple language, because I will forward it to the cooperative board.

We must be able to defend every decision in front of the cooperative leaders. If you remove or change any data, write down what you did and why.

Thank you, Mr. Habimana

The five business questions (attached to the message)

The prepared dataset must make it possible to answer these questions later. Use them to decide which data you need and which features to create.

No.	Question from the cooperatives	Why it matters	What the data must allow
A	How much maize can a farm expect to harvest per hectare?	Plan storage, transport and sales	A reliable yield value for each farm
B	Do fertiliser and improved seed really increase yield?	Decide what inputs to request	Fertiliser compared fairly between farms of different size; correct seed variety
C	How do rainfall, temperature, altitude and irrigation affect yield?	Understand risk from weather and location	Valid weather and altitude values; irrigation recorded clearly
D	Do results differ between provinces, districts and seasons?	Target support where it is needed	One consistent label per district and season; correct planting dates
E	Which farms are at high risk of a very low harvest?	Support farmers early, before harvest	Only information known before harvest, with no leakage of the result

Field notes from the MCSU agronomist

Use this background knowledge to judge whether a value is realistic. A value outside the normal range is not automatically wrong, but you must check it and explain your decision.

- 1 hectare (ha) = 10,000 square metres (m^2).
- Maize grows best at average temperatures of roughly 18 to 27 °C. Farms in the cooler highlands are at higher altitude than farms in the warmer lowlands.
- Soil pH in farmland is normally between about 4 and 8. Maize prefers slightly acidic soil, about pH 5.5 to 7.
- Typical smallholder yields are often between about 1,000 and 4,000 kg per hectare; improved or hybrid seed with good fertiliser use can give more. A yield far outside this range, or a negative yield, must be checked before it is trusted.
- The fertiliser amount in the file is the total applied on the whole farm, not the amount per hectare.
- Irrigation is mostly used in Season C. In Seasons A and B most plots depend on rain.
- All farmers in the file are adults.

Rules from your manager

- Never edit the original file. Keep it in data/raw/ and do every change in code, so that the work can be repeated from the raw file to the final dataset.
- Do not build or train a prediction model. Your job ends with a model-ready dataset.
- No personal information may appear in your outputs. Farm and cooperative codes are allowed.
- Write down every decision (what, why, how many records affected) in the Cleaning Log.
- Finish before the end of Session 3. Late work delays the modelling team.

What you must deliver

No.	Deliverable	Stage	Criterion	Evidence
1	Working environment and a Git repository that anybody can use to repeat your work	1	1.1	Repository, environment.yml, README
2	Data requirements written from the manager's message and the five questions, a check that the file meets them, and a completed data dictionary	2	1.2	Requirements, dictionary, source log
3	Data Quality Report: every problem found, the columns and number of records affected, how serious it is (High / Medium / Low) and the proposed treatment. At least eight different kinds of problems are hidden in the file	3	1.3	Notebook 02_eda, report
4	Clean dataset, Cleaning Log, reusable pipeline, and a train / validation / test split without data leakage	4	1.3	Notebook 03_clean, log, data files
5	At least five engineered features and a Feature Justification table linked to the five business questions	5	1.4	Notebook 04_features, table
6	One-page memo to Mr. Habimana (template in Appendix E)	6	1.1 to 1.4	Memo

Definition of done: the dataset is ready when

Check	Standard to reach
Duplicates	No repeated farm records remain
Data types	Every column has the correct type (numbers are numbers, dates are dates)
Consistency	One unit for farm size and one label for each category, season and district
Realistic values	Values outside the normal range are checked and handled, with a reason
Missing values	Each column with missing or disguised-missing values has a justified treatment

Check	Standard to reach
No leakage	No column that would only be known after harvest is used to describe the farm before harvest
Split	Train / validation / test files saved with a fixed random seed; scaling and imputing learned from the training set only
Features	At least five new features, each one explained
Repeatable	Running the notebooks in order on the raw file gives the same final dataset

Data supplied for Question 1

File: maize_yield_rwanda_raw.csv (about 1,200 records and 20 columns). The target to be predicted later in Element 2 is yield_kg_per_ha.

Column	Meaning	Unit / type
farm_id	Farm record code	text
province, district	Location of the farm	category
cooperative_id	Cooperative of the farmer (20 cooperatives)	category
altitude_m	Altitude of the farm	metres
season	Agricultural season (A, B or C)	category
planting_date	Date of planting	date
farm_size, farm_size_unit	Size of the farm and the unit in which it was recorded	number, text
seed_variety	Local, Improved or Hybrid	category
fertilizer_kg	Total fertiliser applied on the whole farm	kg
rainfall_mm	Rainfall during the season	mm
avg_temp_c	Average temperature during the season	°C
soil_type, soil_ph	Soil type (Clay, Loam, Sandy) and soil acidity	category, pH
pest_incidence	Level of pests (Low, Medium, High)	category
irrigation	Whether the farm was irrigated	Yes / No

Column	Meaning	Unit / type
farmer_age	Age of the farmer	years
harvest_weight_kg	Total harvest recorded for the farm	kg
yield_kg_per_ha	Maize yield per hectare (the target)	kg per hectare

Question 2: Create Your Own Scenario

Now it is your turn to be the data scientist and the writer. Create a scenario in the same style as Question 1, then prepare the data for it by following the same stages. Your scenario must meet these requirements:

- It is set in Rwanda: name the district or sector, and the field (agriculture, health, education, transport, energy, finance, tourism, environment, public services, etc.).
- It has a realistic stakeholder (a cooperative, health centre, school, SACCO or bank, telecom, city or district office, etc.) who has a decision to make that a prediction could support.
- It has a clear target variable to be predicted later, and you explain why predicting it matters.
- It is in a different field or problem from Question 1 and from the scenarios of other groups.
- It is written as a short scenario brief of half a page to one page: who is asking, what the problem is, what data exists, and what the manager asks you to deliver.
- The data you find or collect meets the minimum dataset requirements below, and your scenario brief is approved before you start (Stage 0).

Question 2 is marked on the same four criteria as Question 1, but your group chooses and justifies everything: the scenario, the data, the cleaning decisions and the features.

Ideas to start from

Possible scenario in Rwanda	Possible target to predict
House or land price in Kigali for a real estate agency	Price (regression)
Student performance at a secondary school or TVET	Pass/fail or final mark
Mobile money fraud for a telecom or bank	Fraud / not fraud (classification)
Air quality or rainfall for a city or district office	Pollution level / rainfall
Customer churn for a telecom or SACCO	Customer leaves / stays

Possible scenario in Rwanda	Possible target to predict
Road traffic accidents for a traffic safety unit	Accident severity
Household electricity consumption for a utility	Monthly consumption (regression)
Disease cases (e.g. malaria) at a district hospital	Number of cases / risk level
Tourist arrivals at a national park or hotel	Number of visitors

Minimum dataset requirements for Question 2

- At least 1,000 rows and at least 8 columns (features), including both numerical and categorical columns.
- A clearly defined target column.
- Open licence or permission to use; no personal identifiers (names, phone numbers, national ID numbers). Anonymise if needed.
- The raw data should contain some real quality problems (missing values, duplicates, outliers, inconsistent text). Data that is already perfectly clean is not accepted.
- Image, audio or text projects are allowed only with approval, because their preparation steps are different.

Possible data sources: Kaggle, UCI Machine Learning Repository, Rwanda Open Data / NISR, World Bank and WHO open data, Zindi competitions, Hugging Face datasets, or data your group collects itself (e.g. with KoboToolbox or Google Forms, after approval).

4. Overview of the Process (from Start to Final Product)

The task is divided into seven stages. You do the stages first for Question 1 (guided, data supplied) and then repeat them for Question 2 (your own scenario). In Question 2 you may reuse and adapt your own Question 1 notebooks, but every decision must be justified again for the new data. Each stage ends with a checkpoint and produces evidence that is stored in the group repository. The evidence for every checkpoint must include the screenshots listed in Section 2.

Stage	Title	Criterion	Applies to	Main output
0	Create your own scenario and proposal	Preparation	Q2 only	Approved scenario brief
1	Configure the environment	1.1	Q1, reused in Q2	Working environment and Git repository
2	Define requirements and acquire data	1.2	Q1 and Q2	Raw dataset, data dictionary, source log
3	Explore and audit the data (EDA)	1.3	Q1 and Q2	Data quality report
4	Clean and pre-process the data	1.3	Q1 and Q2	Clean dataset and train/validation/test split
5	Engineer and select features	1.4	Q1 and Q2	Feature table and feature justification
6	Finalise, package and present	1.1 - 1.4	Q1 and Q2	Model-ready datasets, pipelines, memo, report, demo

5. Detailed Stages

Stage 0: Create Your Own Scenario and Proposal Question 2 only	
Objective	Invent a realistic Rwandan scenario, confirm that suitable data exists, and get the scenario approved.
What you do	Form a group of two and register the names. Brainstorm 2-3 scenario ideas, search quickly for data for each, and choose one. Write the scenario brief (half a page to one page) in the style of Question 1: who is asking, the problem, the data, and what the manager wants delivered. Complete the Scenario Proposal Form (Appendix A): location, stakeholder, decision to support, target variable, expected data source, expected size and risks.
Evidence to submit	Scenario brief and signed Scenario Proposal Form.

Checkpoint	Scenario approved (signature in Appendix A) before Stage 2 of Question 2 starts. No two groups may have the same scenario or dataset.

Stage 1: Configure the Environment PC 1.1	
Objective	Prepare a working and reproducible environment that matches what the project needs. This is done once at the start of Question 1 and reused in Question 2.
What you do	Install Anaconda (Python 3.10 or newer) and open Jupyter Notebook from Anaconda Navigator. Both members must have a working installation on their own computer. Create a new conda environment for the project (for example: conda create -n ml-project python=3.10) and activate it in Anaconda Prompt. Install the libraries in that environment: pandas, numpy, matplotlib, seaborn, scikit-learn, and others if the project needs them (e.g. openpyxl, requests). Export the environment to environment.yml (conda env export > environment.yml) so that the work can be repeated on another computer. Create a Git repository (GitHub/GitLab) with the folder structure shown below; both members clone it and make at least one commit each. Start Jupyter Notebook in the project folder. Write and run a short test notebook that imports all libraries and prints their versions. Take the screenshots required for this stage (see Section 2). Explain in the README why each tool and library was chosen for this project.
Evidence to submit	Repository link with commits from both members. environment.yml and README.md. Screenshots: Anaconda Navigator, conda environment list, library version test.
Checkpoint	Each student runs the test notebook on their own computer and shows the output.

Suggested repository structure

```
group-XX-repository/
  environment.yml   README.md
  q1_maize_yield/
    data/raw/        (original data, never edited)
    data/interim/    (data being cleaned)
    data/processed/  (final model-ready data)
    notebooks/       (01_acquire, 02_eda, 03_clean, 04_features)
    docs/            (dictionary, logs, memo)
  q2_own_scenario/   (same structure as q1_maize_yield)
```

Stage 2: Define Requirements and Acquire Data PC 1.2	
Objective	Turn the problem into clear data requirements, then obtain the data from a legitimate source and load it correctly.
Q1 / Q2	Question 1: you are given the file maize_yield_rwanda_raw.csv. You still write the data requirements from Mr. Habimana's message and the five business questions, check that the file meets them, and complete the dictionary and source log. Question 2: you search for or collect the data yourselves and justify the source.
What you do	Write the data requirements: what is the target, which features could influence it, what time period, what area, how many records are needed. (Question 2) Search and compare at least two possible sources; choose one and record why (relevance, size, licence, quality). Acquire the data (Question 1: copy the supplied file; Question 2: download, API, web scraping where allowed, survey or merge of several files) and keep the original files untouched in data/raw/. Load the data into a pandas DataFrame; check the number of rows and columns, data types and the first and last records. Complete the Data Dictionary (Appendix B): column name, meaning, type, unit, allowed values. Complete the Source Log: source, link, date accessed, licence, who collected it, limitations and ethical issues (privacy, bias).
Evidence to submit	Raw data in data/raw/ and notebook 01_acquire. Data requirements statement, Data Dictionary and Source Log.
Checkpoint	The data matches the requirements and the licence allows use (confirmed at the checkpoint).

Stage 3: Explore and Audit the Data (EDA) PC 1.3	
Objective	Understand the data and find every quality problem before changing anything.
What you do	Describe the dataset: shape, data types, summary statistics (describe, info), unique values of categorical columns. Count missing values per column, duplicated rows, impossible values (e.g. negative age), and inconsistent spellings or formats. Detect outliers with box plots and the IQR rule (or z-score). Study the target: distribution, and class balance if it is a classification problem. Draw at least five meaningful charts (histograms, bar charts, scatter plots, correlation heatmap) and write one sentence of interpretation under each. Produce a Data Quality Report: a table listing each problem found, the affected columns, how many records, and the proposed treatment.
Evidence to submit	Notebook 02_eda with charts and interpretations. Data Quality Report (docs/).
Checkpoint	Both students explain two problems found and why they matter for the model.

Stage 4: Clean and Pre-process the Data PC 1.3	
Objective	Transform the raw data into a clean dataset that suits the intended model, without leaking information.
What you do	Remove or merge duplicates; correct wrong data types, formats, spellings and units. Handle missing values: delete, or impute with mean / median / mode / group value / model-based value. Justify the choice for each column. Handle outliers: keep, cap (winsorise), transform or remove, with a reason. Encode categorical variables: label/ordinal encoding or one-hot encoding as appropriate. Scale numerical variables (standardisation or min-max) where the algorithm needs it. If the target is imbalanced, record the imbalance and the planned remedy (e.g. class weights, resampling on training data only). Split into training, validation and test sets (for example 70/15/15) with a fixed random seed. Fit scalers and imputers on the training set only. Keep a Cleaning Log: every operation, number of records affected, before and after values. Save the clean data in data/interim/ and put the steps in a reusable scikit-learn Pipeline or a documented function.

Evidence to submit	Notebook 03_clean. Cleaning Log (docs/). Clean dataset and train/validation/test files.
Checkpoint	Re-running the notebook from top to bottom gives the same clean dataset. Each student answers one question on data leakage.

Stage 5: Engineer and Select Features PC 1.4	
Objective	Create and choose the features that carry the most useful information for the problem, based on what the data analysis showed.
What you do	Create at least five new features from existing ones, for example ratios, differences, date parts (month, weekday), counts, group averages, binning of continuous variables, interaction terms. Apply transformations where the analysis shows skewness (log, square root) and explain why. Check relationships: correlation matrix, mutual information or feature importance (e.g. a quick Random Forest) against the target. Select features: remove constant, duplicated and highly correlated features, and use at least one selection method (variance threshold, SelectKBest, or model-based). Optional extension: reduce dimensions with PCA and compare with the original feature set. Write a Feature Justification table: feature, how it was created or why it was kept or removed, evidence from the analysis. Make sure the same feature steps are applied to training, validation and test data.
Evidence to submit	Notebook 04_features. Feature Justification table. Final feature set in data/processed/.
Checkpoint	Each student explains the origin and purpose of three features of their choice.

Stage 6: Finalise, Package and Present PC 1.1 - 1.4	
Objective	Deliver a complete, reproducible product that the next element (Apply Machine Learning techniques) can use immediately.
What you do	Save the final train, validation and test files (CSV or parquet) in data/processed/. Create a Data Card (one page): purpose, source, size, columns, target, cleaning and feature steps, known limitations and bias risks, how to reproduce. Check that someone else can clone the repository, create the Anaconda environment from environment.yml, run the notebooks in order and get the same final dataset. Write the one-page memo to the manager (Question 1) and complete the documentation report with screenshots for both questions (see Sections 2 and 6) and the Contribution Log (Appendix D). Prepare your presentation for Thursday 12 November 2026 (morning): 10 minutes covering both questions, followed by 5 minutes of questions.
Evidence to submit	Final repository link. Data Card, documentation report with screenshots and Contribution Log. Demo slides (optional).
Checkpoint	Presentation on Thursday 12 November 2026 and individual oral questions.

6. Final Deliverables

No.	Deliverable	Content
1	Git repository	One repository with folders q1_maize_yield and q2_own_scenario (raw data, notebooks 01-04, processed data), environment.yml, README, commits from both members
2	Model-ready datasets	For Q1 and Q2: train, validation and test files with final features and a clearly named target
3	Reusable pre-processing pipelines	For Q1 and Q2: pipeline or functions that repeat all cleaning and feature steps on new data
4	Documentation packs	For Q1 and Q2: Data Dictionary, Source Log, Data Quality Report, Cleaning Log, Feature Justification, Data Card
5	Memo to the manager (Q1)	One page in simple language: usable records, problems found, main decisions, new features, limits of the data
6	Scenario brief (Q2)	Half a page to one page, approved at Stage 0

No.	Deliverable	Content
7	Documentation report with screenshots	Both questions: problem, environment, data sources, quality problems, cleaning decisions, features, limitations, lessons learned. About 6 to 10 pages of text plus the screenshots required in Section 2
8	Presentation	Thursday 12 November 2026 (morning): 10 minutes covering both questions + 5 minutes of questions
9	Contribution Log	What each member did in each stage, role rotation, signed by both students

7. Suggested Timeline

All work must be finished and ready for the presentation on Thursday 12 November 2026 (morning). The six sessions below must be completed before that date, so plan time outside class for data search, scenario writing and documentation with screenshots.

Session	Work	Checkpoint
1	Briefing. Q1: Stage 1 (environment) and start of Stage 2 (requirements, load the data)	Environment test
2	Q1: finish Stage 2, Stage 3 (EDA) and start of Stage 4 (cleaning)	Data Quality Report
3	Q1: finish Stage 4, Stage 5 (features) and the manager memo. Q2: Stage 0 (scenario brief)	Q1 clean dataset; Q2 scenario approved
4	Q2: Stage 2 (requirements and data acquisition) and Stage 3 (EDA)	Q2 data and quality report
5	Q2: Stage 4 (cleaning) and Stage 5 (features)	Q2 notebooks run end to end
6	Stage 6: packaging, documentation with screenshots and presentation practice (both questions)	Presentation on Thursday 12 November 2026

8. Assessment Rubric (100 Marks)

Criterion	Q1	Q2	Total	What the assessor looks for
1.1 Environment properly configured	5	10	15	Working Anaconda environment, correct libraries and versions, environment.yml, Git repository with commits from both members, choice of tools justified

Criterion	Q1	Q2	Total	What the assessor looks for
1.2 Data properly acquired	5	15	20	Data matches the requirements, clear requirements statement, legitimate source and licence, data dictionary, source log, original data kept unchanged. For Q2 also: a realistic, clearly written Rwandan scenario
1.3 Data appropriately pre-processed	15	15	30	Complete quality audit (Q1: all hidden problems found, including leakage), justified handling of missing values, duplicates and outliers, correct encoding and scaling, split without leakage, cleaning log, repeatable pipeline
1.4 Features properly engineered	8	12	20	At least five meaningful new features, appropriate transformations, selection method applied and justified, feature table consistent across all splits
Documentation and reproducibility	2	3	5	Data card, README, manager memo (Q1), notebook that runs from top to bottom, clear comments, screenshots with captions and explanations
Teamwork, demo and oral questions	5	5	10	Equal contribution, role rotation, clear demo, each student answers questions correctly
TOTAL	40	60	100	

Individual adjustment. If the oral questions show that one member did not understand or contribute to the work, that member's marks are adjusted downwards individually, even if the group product is good.

9. Rules and Expectations

- Your Question 2 scenario and dataset must differ from those of other groups and be approved before you start. The Question 1 file is for this course only.
- Original raw data is never edited; every change is done in code and recorded.
- Do not use the test set for any decision during cleaning, feature creation or selection.
- Use AI tools or online help only to learn and to debug; you must be able to explain every line you submit. Copying another group's work is not allowed.
- Respect data privacy and licences. Remove or anonymise personal information.
- Submit on time; late checkpoints reduce the marks of the affected criterion.

Appendix A: Question 2 Scenario Proposal Form

Field	Group entry
Group number and members	
Scenario title	
Location in Rwanda (district / sector)	
Stakeholder (who asks for the work)	
Scenario story and decision to support (5-8 sentences)	
Who will benefit from a solution?	
Target variable (what will be predicted)	
Problem type (regression / classification / other)	
Expected data source and link	
Expected number of rows and columns	
Known data risks (privacy, licence, missing data)	
Approval	Approved / Needs changes Signature: _____ Date: _____

Appendix B: Data Dictionary Template

Column name	Meaning	Data type	Unit	Allowed values / notes

Appendix C: Checkpoint Sign-off Sheet

Stage	Evidence checked	Q1 result	Q2 result	Signature
0	Scenario brief and proposal	n/a	OK / Redo	
1	Environment and repository	OK / Redo	reused	
2	Data, dictionary and source log	OK / Redo	OK / Redo	
3	EDA and Data Quality Report	OK / Redo	OK / Redo	
4	Cleaning log and pipeline	OK / Redo	OK / Redo	
5	Features and justification	OK / Redo	OK / Redo	
6	Final product, memo and demo	OK / Redo	OK / Redo	

Appendix D: Individual Contribution Log

Each group completes one log. Record who was Driver and who was Navigator in each stage, and what each student did.

Stage	Driver	Navigator	Student 1 contribution	Student 2 contribution
1				
2				
3				
4				
5				
6				

Student 1 signature: _____ Student 2 signature: _____

Appendix E: Memo Template for Question 1

Write the memo to Mr. Habimana in simple language, on one page, without code. He will forward it to the cooperative board.

Section	What to write
To / From / Date / Subject	To: Mr. Habimana, Head of Planning and Data. From: names of the two analysts. Subject: Maize yield data: ready for modelling?
1. What we received	How many records and columns the file had, and how many records can be used after cleaning
2. Main problems found	The three most serious problems, why each one matters for the cooperatives, and what you did about each
3. Data removed or changed	Which columns or records you removed or changed, and why. Mention any column removed to avoid leakage
4. New features	The most useful new features and how they help to answer the five business questions
5. Limits and risks	What the data still cannot tell us (for example missing information or weak coverage) and the risk of using it
6. Recommendation	Is the data ready for the modelling team? Which of the questions A to E can it support now? What should be collected better next season?